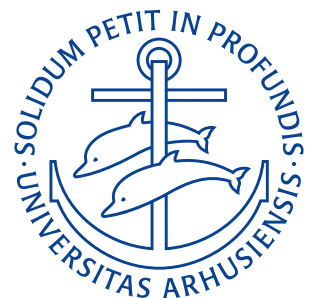


Department of Mechanical and Production Engineering
Aarhus University

PDE Exam 2025

Course: Partial Differential Equations — 290211U002
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Problem 1

Show that

$$f(x) = xe^{\cos x}$$

is an odd function and $f'(x)$ is an even function.

Derivation

For an even function

$$f(-x) = f(x)$$

for all x , whereas for an odd function

$$f(-x) = -f(x)$$

for all x . Therefore, to determine if the functions are even or odd we will try inserting $-x$ as our function argument.

Starting with $f(x)$ we get:

$$\begin{aligned} f(x) &= xe^{\cos x} \\ f(-x) &= (-x)e^{\cos(-x)} \\ &= -xe^{\cos x} \\ &= -f(x). \end{aligned}$$

As $f(-x) = -f(x)$ for all x , $f(x)$ is odd.

Moving on to $f'(x)$ we get

$$\begin{aligned} f'(x) &= e^{\cos x} - x \sin(x) e^{\cos x} \\ f'(-x) &= e^{\cos(-x)} - (-x) \sin(-x) e^{\cos(-x)} \\ &= e^{\cos x} - x \sin(x) e^{\cos x} \\ &= f'(x). \end{aligned}$$

As $f'(-x) = f'(x)$ for all x , $f'(x)$ is even.

$f(x):$ Odd $f'(x):$ Even.

RESULT

Problem 2

Calculate the left-hand derivative and the right-hand derivative of

$$f(x) = (x - 1)|x - 1|$$

at $x = 1$.

Derivation

The left- and right-hand derivatives at x_0 are defined as

$$f'(x_0-) = \lim_{h \rightarrow 0} \frac{f(x_0) - f(x_0 - h)}{h}$$

$$f'(x_0+) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

respectively.

Starting with the left-hand derivative we get

$$\begin{aligned} f'(1-) &= \lim_{h \rightarrow 0} \frac{f(1) - f(1 - h)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1 - 1)|1 - 1| - (1 - h - 1)|1 - h - 1|}{h} \\ &= \lim_{h \rightarrow 0} \frac{0 - (-h)|-h|}{h} \\ &= \lim_{h \rightarrow 0} \frac{h|h|}{h} \\ &= \lim_{h \rightarrow 0} |h| \\ &= 0. \end{aligned}$$

Moving on to the right-hand derivative we get

$$\begin{aligned} f'(1+) &= \lim_{h \rightarrow 0} \frac{f(1 + h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1 + h - 1)|1 + h - 1| - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{h|h|}{h} \\ &= \lim_{h \rightarrow 0} |h| \\ &= 0. \end{aligned}$$

$\begin{aligned} f'(1-) &= 0 \\ f'(1+) &= 0. \end{aligned}$

RESULT

Problem 3

Find a period and the corresponding Fourier coefficients of

$$f(x) = \cos\left(\frac{2\pi}{5}\right) \sin\left(\frac{2x}{\sqrt{5}}\right) + 7 \cos(\sqrt{5}x), \quad -\infty < x < \infty.$$

Derivation

The general representation of a Fourier series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right).$$

We can rewrite the given function as

$$\begin{aligned} f(x) &= 7 \cos\left(\frac{\sqrt{5}\pi x}{\pi}\right) + \cos\left(\frac{2\pi}{5}\right) \sin\left(\frac{2\pi x}{\sqrt{5}\pi}\right) \\ &= \underbrace{7}_{a_5} \cos\left(\underbrace{\frac{5\pi x}{\sqrt{5}\pi}}_L\right) + \underbrace{\cos\left(\frac{2\pi}{5}\right)}_{b_2} \sin\left(\underbrace{\frac{2\pi x}{\sqrt{5}\pi}}_L\right). \end{aligned}$$

Hence, $p = 2L = 2\sqrt{5}\pi$, $a_5 = 7$, $b_2 = \cos(2\pi/5)$, and all other Fourier coefficients are zero.

$p = 2\sqrt{5}\pi \quad a_5 = 7 \quad b_2 = \cos\left(\frac{2\pi}{5}\right).$

RESULT

Problem 4

Consider the function

$$f(x) = \cos\left(\frac{x}{2}\right), \quad 0 \leq x \leq 4\pi.$$

- Find the period and the corresponding Fourier coefficients of the expansion of $f(x)$.
- Find the period and the corresponding Fourier coefficients of the even expansion of $f(x)$.

Derivation

- The expansion of $f(x)$ is

$$f_0(x) = \begin{cases} \cos \frac{x}{2} & 0 \leq x \leq 2\pi \\ -\sin \frac{x}{2} & -2\pi \leq x \leq 0. \end{cases}$$

with period $p = 2\pi$. To find the Fourier coefficients we use the formulae for the Fourier coefficients as:

$$\begin{aligned} a_0 &= \frac{1}{2L} \int_{-L}^L f(x) dx \\ &= \frac{1}{4\pi} \left(\int_0^{2\pi} \cos \frac{x}{2} dx - \int_{-2\pi}^0 \sin \frac{x}{2} dx \right) \\ &= \frac{1}{4\pi} \cdot (0 - (-4)) \\ &= \frac{4}{4\pi} \\ &= \frac{1}{\pi}. \end{aligned}$$

We now move on to a_n as

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{2\pi} \left(\int_0^{2\pi} \cos\left(\frac{x}{2}\right) \cos\left(\frac{n\pi x}{2\pi}\right) dx - \int_{-2\pi}^0 \sin\left(\frac{x}{2}\right) \cos\left(\frac{n\pi x}{L}\right) dx \right) \\ &= \frac{1}{2\pi} \left(\frac{2(\cos(\pi n) + 1)}{n^2 - 1} - \frac{2n \sin(\pi n)}{n^2 - 1} \right) \\ &= \frac{2(\cos(\pi n) - n \sin(\pi n) + 1)}{2\pi(n^2 - 1)}. \end{aligned}$$

And b_n is found as

$$\begin{aligned} b_n &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{1}{2\pi} \left(\int_0^{2\pi} \cos\left(\frac{x}{2}\right) \sin\left(\frac{n\pi x}{2\pi}\right) dx - \int_{-2\pi}^0 \sin\left(\frac{x}{2}\right) \sin\left(\frac{n\pi x}{L}\right) dx \right) \\ &= \frac{1}{2\pi} \left(\frac{2n \cos(n\pi) + 1}{n^2 - 1} - \frac{2 \sin(\pi n)}{n^2 - 1} \right) \\ &= \frac{2(n \cos(n\pi) - \sin(\pi n) + 1)}{2\pi(n^2 - 1)}. \end{aligned}$$

I.e. the period $p^0 = 2L = 2\pi$ and the Fourier coefficients are

$$\begin{aligned} a_0^0 &= \frac{1}{\pi} \\ a_n^0 &= \frac{2(\cos(\pi n) - n \sin(\pi n) + 1)}{2\pi(n^2 - 1)} \end{aligned}$$

$$b_n^0 = \frac{2(n \cos(n\pi) - \sin(n\pi)) + 1}{2\pi(n^2 - 1)}.$$

b) We now move on to the even expansion, given by

$$f(x) = \cos \frac{x}{2}, -4\pi < x < 4\pi.$$

As this is an even function we must find a Fourier cosine series. We compare with the general form of Fourier cosine series and get:

$$f(x) = \cos \frac{\pi x}{2\pi}$$

$$f(x)^1 = a_0^1 + \sum_{n=1}^{\infty} a_n^1 \cos\left(\frac{n\pi x}{L}\right)$$

And see that $p = 2L = 4\pi$, $a_1^1 = 1$, and all other Fourier coefficients are zero.

a) $p^0 = 2\pi, \quad a_0^0 = \frac{1}{\pi},$ $a_n^0 = \frac{2(\cos(\pi n) - n \sin(\pi n) + 1)}{2\pi(n^2 - 1)},$ $b_n^0 = \frac{2(n \cos(\pi n) - \sin(\pi n)) + 1}{2\pi(n^2 - 1)}.$ b) $p = 4\pi, \quad a_1^1 = 1.$

RESULT

Problem 5

Rewrite the PDE

$$u_{yy}(x, y) - u_{xx}(x, y) = x$$

in terms of the variables

$$p(x, y) = x + y, \quad q(x, y) = x - y.$$

You may assume that you can change the order of derivatives in mixed second derivatives.

Derivation

We define

$$\bar{u}(p, q) = \bar{u}(p(x, y), q(x, y)) = u(x, y) \implies \bar{u}_{yy}(p, q) - \bar{u}_{xx}(p, q) = p + q.$$

The first derivative with respect to x is

$$\begin{aligned} \bar{u}_x(p, q) &= \bar{u}_p(p, q)p_x(x, y) + \bar{u}_q(p, q)q_x(x, y) \\ &= \bar{u}_p(p, q) \cdot 1 + \bar{u}_q(p, q) \cdot 1 \\ &= \bar{u}_p(p, q) + \bar{u}_q(p, q). \end{aligned}$$

The second derivative with respect to x is

$$\begin{aligned} \bar{u}_{xx}(p, q) &= (\bar{u}_p(p, q) + \bar{u}_q(p, q))_x \\ &= (\bar{u}_p(p, q))_x + (\bar{u}_q(p, q))_x \\ &= \bar{u}_{pp}(p, q)p_x(x, y) + \bar{u}_{pq}(p, q)q_x(x, y) + \bar{u}_{qp}(p, q)p_x(x, y) + \bar{u}_{qq}(p, q)q_x(x, y) \\ &= \bar{u}_{pp}(p, q) + 2\bar{u}_{pq}(p, q) + \bar{u}_{qq}(p, q). \end{aligned}$$

The first derivative with respect to y is

$$\begin{aligned} \bar{u}_y(p, q) &= \bar{u}_p(p, q)p_y(x, y) + \bar{u}_q(p, q)q_y(x, y) \\ &= \bar{u}_p(p, q) \cdot 1 + \bar{u}_q(p, q) \cdot (-1) \\ &= \bar{u}_p(p, q) - \bar{u}_q(p, q). \end{aligned}$$

The second derivative with respect to y is

$$\begin{aligned} \bar{u}_{yy}(p, q) &= (\bar{u}_p(p, q) - \bar{u}_q(p, q))_y \\ &= (\bar{u}_p(p, q))_y - (\bar{u}_q(p, q))_y \\ &= \bar{u}_{pp}(p, q)p_y(x, y) + \bar{u}_{pq}(p, q)q_y(x, y) - \bar{u}_{qp}(p, q)p_y(x, y) - \bar{u}_{qq}(p, q)q_y(x, y) \\ &= \bar{u}_{pp}(p, q) - 2\bar{u}_{pq}(p, q) + \bar{u}_{qq}(p, q). \end{aligned}$$

By insertion we get

$$\begin{aligned} \bar{u}_{pp}(p, q) - 2\bar{u}_{pq}(p, q) + \bar{u}_{qq}(p, q) - \bar{u}_{pp}(p, q) + 2\bar{u}_{pq}(p, q) + \bar{u}_{qq}(p, q) &= p + q \\ 2\bar{u}_{qq}(p, q) &= p + q. \end{aligned}$$

$2\bar{u}_{qq}(p, q) = p + q$

RESULT

Problem 6

Consider the PDE

$$u_{xx}(x, t) + u_{tt}(x, t) = u_t(x, t) + u(x, t), \quad -\infty < x < \infty, \quad t \geq 0$$

and let $\hat{u}(\omega, t) = \mathcal{F}(u(x, t))$ be the Fourier transform of $u(x, t)$ with respect to x . Derive an ODE for $\hat{u}(\omega, t)$ with respect to t . You may assume that the Fourier transforms of $u(x, t)$, $u_t(x, t)$, $u_{xx}(x, t)$, and $u_{tt}(x, t)$ exist and that

$$\mathcal{F}(u_t(x, t)) = \frac{\partial}{\partial t} \mathcal{F}(u(x, t)), \quad \mathcal{F}(u_{tt}(x, t)) = \frac{\partial^2}{\partial t^2} \mathcal{F}(u(x, t)).$$

Derivation

We know that $\mathcal{F}(f'') = -\omega^2 \mathcal{F}(f)$. Hence, we get

$$\begin{aligned} \mathcal{F}(u_{xx}) + \mathcal{F}(u_{tt}) &= \mathcal{F}(u_t) + \mathcal{F}(u) \\ -\omega^2 \mathcal{F}(u) + \frac{\partial^2}{\partial t^2} \mathcal{F}(u) &= \frac{\partial}{\partial t} \mathcal{F}(u) + \mathcal{F}(u) \\ -\omega^2 \hat{u}(\omega, t) + \frac{\partial^2}{\partial t^2} \hat{u}(\omega, t) &= \frac{\partial}{\partial t} \hat{u}(\omega, t) + \hat{u}(\omega, t). \end{aligned}$$

$$-\omega^2 \hat{u}(\omega, t) + \frac{\partial^2}{\partial t^2} \hat{u}(\omega, t) = \frac{\partial}{\partial t} \hat{u}(\omega, t) + \hat{u}(\omega, t)$$

RESULT

Problem 7

Consider the deflection $u(x, y, t)$ of a vibrating elastic membrane in the xy -plane, $0 \leq x \leq 1$, $0 \leq y \leq 1$, fixed along its boundaries and governed by

$$u_{tt}(x, y, t) = (u_{xx}(x, y, t) + u_{yy}(x, y, t))$$

with general solution

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (E_{m,n} \cos(\lambda_{m,n} t) + J_{m,n} \sin(\lambda_{m,n} t)) \sin(m\pi x) \sin(n\pi y)$$

and

$$\lambda_{m,n} = \pi \sqrt{m^2 + n^2}.$$

Find the particular solution that corresponds to the initial conditions

$$u(x, y, 0) = \sin(\pi x) \sin(\pi y) + \sin(2\pi x) \sin(2\pi y)$$

and

$$u_t(x, y, 0) = 0.$$

Derivation

We employ our initial elongation as:

$$\begin{aligned} u(x, y, 0) &= \sin(\pi x) \sin(\pi y) + \sin(2\pi x) \sin(2\pi y) \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} E_{m,n} \sin(m\pi x) \sin(n\pi y) \\ \implies E_{1,1} &= 1, \quad E_{2,2} = 1. \end{aligned}$$

Using the initial velocity we get

$$\begin{aligned} u_t(x, y, 0) &= 0 \\ &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (J_{m,n} \lambda_{m,n} \cos(\lambda_{m,n} t)) \sin(m\pi x) \sin(n\pi y) \\ \implies J_{m,n} &= 0. \end{aligned}$$

We know that

$$\lambda_{m,n} = \pi \sqrt{m^2 + n^2}.$$

Hence, $\lambda_{1,1}$ and $\lambda_{2,2}$ are found as:

$$\begin{aligned} \lambda_{1,1} &= \pi\sqrt{2} \\ \lambda_{2,2} &= \pi\sqrt{8}. \end{aligned}$$

$$u(x, y, t) = \cos(\pi\sqrt{2}t) \sin(\pi x) \sin(\pi y) + \cos(\pi\sqrt{8}t) \sin(2\pi x) \sin(2\pi y)$$

RESULT

Problem 8

Consider the PDE

$$u_{xx}(x, y, t) + 3u_y(x, y, t) - u_t(x, y, t) = u(x, y, t).$$

Use the method of separation of variables to convert the above PDE into three ODEs.

Derivation

Using the method of separation of variables we assume $u(x, y, t) = F(x, y)G(t)$ to first derive a PDE for $F(x, y)$ and an ODE for $G(t)$ as

$$\begin{aligned} F_{xx}(x, y)G(t) + 3F_y(x, y)G(t) - F(x, y)G'(t) &= F(x, y)G(t) \\ \frac{F_{xx}(x, y) + 3F_y(x, y) - F(x, y)}{F(x, y)} &= \frac{G'(t)}{G(t)} = c_0 \\ \implies G'(t) - c_0G(t) &= 0 \\ F_{xx}(x, y) + 3F_y(x, y) - F(x, y) - c_0F(x, y) &= 0. \end{aligned}$$

We now assume $F(x, y) = H(x)Q(y)$ and get

$$\begin{aligned} H''(x)Q(y) + 3H(x)Q'(y) - H(x)Q(y) - c_0H(x)Q(y) &= c_1 \\ \frac{H''(x)}{H(x)} - \frac{3Q'(y)}{Q(y)} &= 1 + c_0 + c_1 = c_2 \\ \implies H''(x) - c_2H(x) &= 0 \\ -3Q'(y) - c_2Q(y) &= 0. \end{aligned}$$

Hence,

$\begin{aligned} H''(x) - c_2H(x) &= 0 \\ -3Q'(y) - c_2Q(y) &= 0 \\ G'(t) - c_0G(t) &= 0. \end{aligned}$
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RESULT